MCQ solutions

1. **C.** d/dx of $x^3/3$ gives back x^2. Differentiate x^2 is not finding an antiderivative.
2. **B.** The factor 1/2 compensates for the factor 2 from differentiating the composite. C is a real constant.
3. **D.** This is the form u'/u, with $u = 1 + x^2 > 0$ everywhere. $\ln x$ is not defined on all $\mathbb{R}$.
4. **A.** The difference between antiderivatives is constant; this translates into a vertical translation.
5. **C.** The antiderivatives are $x^2 + C$. The condition gives $1 + C = 3$, so $C = 2$.
6. **B.** The solutions are Ce^{-2t}; the initial condition gives $C = 3$. An affine function is not suitable at every instant.

7. **D.** A constant has zero derivative; solve $0 = -2L + 6$ gives $L = 3$.
8. **A.** Subtracting the two equations eliminates the right-hand side. This is the associated homogeneous equation.
9. **C.** The update is $y_1 = y_0 + h(-y_0) = 2 - 2h$. Euler gives an approximation, distinct from the exact solution $2e^{-h}$.
10. **B.** The derivative is in degrees Celsius per minute: k multiplies a temperature difference and has unit min^{-1}.

True/false. False: adding a constant gives another antiderivative. False: for $y' = -y$, $y(0) = 2$, a step of 1 gives 0, instead of $2/e$. True, on the required interval and with the required differentiability.

Solutions to the applications and guidance

Application 1. $x^4 - x^2 + x$; $-e^{-3x}/3$; $\frac{1}{2}\ln|2x+1|$ on each of the intervals not containing $-1/2$. Differentiating gives back the stated expressions.

Application 2. $F = x^2 + C$ and $F(1) = 3$ give C=2. On the two half-lines, $\ln|x| + C_-$ for $x < 0$ and $\ln|x| + C_+$ for $x > 0$ is an antiderivative, even if $C_- \neq C_+$.

Application 3. $y = 5e^{-2t}$. The zero function satisfies the equation alone, but not the condition $y(0) = 5$.

Application 4. The constant 2 is a solution. $z = y - 2$ satisfies $z' = 2z$ and $z(0) = 1$, therefore $y = 2 + e^{2t}$. Its derivative $2e^{2t}$ is indeed 2 $y - 4$ and $y(0) = 3$.

Application 5. Checking e^t only shows that it is a solution. The functions 2 e^t, $-e^t$ and the zero function are others. For completeness, start from an arbitrary solution and show that it has the stated form.

Application 6. Continuous: $e^{-0{,}1t} = 1/2$, i.e. $t = 10\ln 2 \simeq 6{,}93$ min. Euler's method: $T_n = 20 + 60(0{,}8)^n$; $(0{,}8)^3 = 0{,}512 > 1/2$, $(0{,}8)^4 = 0{,}4096 < 1/2$. The first grid point is therefore at 8 min. The step imposes discrete times and Euler's method introduces an approximation error.

Research and comparison

Have students calculate a first step by hand, then compare h=2 and h=1 with the given program.
The observed convergence of an approximation does not replace an estimate of its error.
Do not call a list of Euler values an "exact solution".

Teaching notes and solutions

Guidance for the teacher

Distinguish the unknown function from the real variable. Have each antiderivative checked by differentiation. For $y' = ay + b$, first seek the constant solution. The case $a = 0$ must remain visible. The trigonometric antiderivatives are given here as a reference table, but their assessed use must wait for the trigonometry unit. For Euler's method, keep the units and compare with a known exact solution.

Solutions to the activity

1. The equation is $T' = -0{,}1T + 2$. A constant solution is 20; all solutions are $T(t) = 20 + Ce^{-0{,}1t}$. Since $T(0) = 80$, $C = 60$. We check $T'(t) = -6e^{-0{,}1t} = -0{,}1(T(t) - 20)$ and $T(0) = 80$.

2. $T'(t) = -6e^{-0{,}1t} < 0$: the temperature strictly decreases. Since $e^{-0{,}1t} \to 0$ at $+\infty$, $T(t) \to 20\,°\text{C}$, remaining above 20 °C.

3. $T(t) = 40 \iff e^{-0{,}1t} = \frac{1}{3}$. Taking the logarithm, $-0{,}1t = -\ln 3$, i.e. $t = 10\ln 3 \simeq 10{,}9861$ min. Strict decrease ensures that this is the only time it passes through 40 °C.

4. For $h = 1$, $Y_{k+1} = Y_k - 0{,}1(Y_k - 20)$. Thus $Y_1 = 80 - 6 = 74$ and $Y_2 = 74 - 5{,}4 = 68{,}6$. The exact solution gives $T(2) = 20 + 60e^{-0{,}2} \simeq 69{,}1238$. Here Euler underestimates the temperature by about $0{,}5238\,°\text{C}$.

5. The model assumes an ambient temperature of 20 °C and a constant exchange coefficient of $0{,}1\ \text{min}^{-1}$. A change of room, a lid or stirring may alter these assumptions; the formula is then no longer automatically appropriate.

E1. Linearity gives $F(x) = x^3 - 2x^2 + x + C$, $C \in \mathbb{R}$. We check $F'(x) = 3x^2 - 4x + 1$. All antiderivatives on the interval $\mathbb{R}$ differ by a constant.

E2. The denominator is positive and its derivative is $2x$. The antiderivatives are $\ln(x^2+4) + C$. The condition at 0 gives $C = -\ln 4$, so $F(x) = \ln(x^2+4) - \ln 4$; we check $F(0) = 0$ and $F'(x) = \frac{2x}{x^2+4}$.

E3. The solutions of $y' = 3y$ are Ce^{3x}. The condition $Ce^3 = 2$ requires $C = 2e^{-3}$. Thus $y(x) = 2e^{3(x-1)}$, whose derivative is $3y(x)$ and whose value at 1 is 2.

E4. The constant 4 is a particular solution. The solutions are therefore $4 + Ce^{-x}$. Since $4 + C = 7$, $C = 3$. The solution is $y(x) = 4 + 3e^{-x}$; its derivative $-3e^{-x}$ is indeed $-y + 4$, and $y(0) = 7$.

E5. $g(x) = x + 1$ satisfies $g' = 1 = g - x$. If $z = y - g$, then $z' = z$, therefore $y = x + 1 + Ce^x$. The condition $y(0) = 0$ requires $C = -1$. We obtain $y = x + 1 - e^x$ and we check $y' = 1 - e^x = y - x$.

Solutions to homework DM7

1. If $g(t) = at + b$, then $g' = a$ and $-g + 2t + 1 = (2 - a)t + 1 - b$. For the identity to hold for every t, we need $2 - a = 0$ and $a = 1 - b$. Hence $a = 2$, $b = -1$, and $g(t) = 2t - 1$.

2. Let $z = y - g$. By subtracting the two equations, $z' = -z$, therefore $z = Ce^{-t}$. Thus $y(t) = 2t - 1 + Ce^{-t}$. Since $y(0) = 3$, $C = 4$. The function $y(t) = 2t - 1 + 4e^{-t}$ does satisfy $y' = 2 - 4e^{-t} = -y + 2t + 1$ and $y(0) = 3$.

3. $y'(t) = 2 - 4e^{-t}$. This derivative is negative for $t < \ln 2$, zero at $\ln 2$, positive after. The solution decreases on $[0, \ln 2]$ then increases on $[\ln 2, +\infty[$. Its minimum is $y(\ln 2) = 2\ln 2 - 1 + 2 = 1 + 2\ln 2$.

4.
```
def euler_dm7(N):
    h = 2/N
    y = 3.0
    for k in range (N):
        t = k*h
        y = y+h*(-y+2*t+1)
    return y
```

The time used on the right-hand side is the time at the start of the step. The program gives $Y_{10} \simeq 3{,}42950$ and $Y_{100} \simeq 3{,}53048$, whereas $y(2) = 3 + 4e^{-2} \simeq 3{,}54134$. The absolute errors are about $0{,}11184$ and $0{,}01086$. To check these values without a loop, let $Z_k = Y_k - (2kh - 1)$. The Euler's recurrence relation gives $Z_{k+1} = (1-h)Z_k$ and $Z_0 = 4$. Hence $Y_N = 3 + 4(1 - \frac{2}{N})^N$.

5. A solution of a differential equation must satisfy an equality between functions on the whole interval, as well as the initial condition. A few numerical values only check a finite number of points; the symbolic substitution in question 2 is a check over the whole domain.

Solutions to the textbook or past-paper work

Exercise 30. If y is constant equal to ℓ, the equation becomes $0 = -\frac{1}{2}\ell + 3$, so $\ell = 6$. Setting $z = y - 6$, we get $z' = -\frac{1}{2}z$; all solutions are $y(x) = 6 + Ce^{-\frac{x}{2}}$, $C \in \mathbb{R}$.

Exercise 25.

1. $f' = \frac{1}{2}f$ gives $f(x) = Ce^{\frac{x}{2}}$. Since $f(0) = 1$, $C = 1$.
2. $5f' = f$ gives $f(x) = Ce^{\frac{x}{5}}$. The condition $f(1) = 0$ requires $Ce^{\frac{1}{5}} = 0$; since the exponential is never zero, $C = 0$. The solution is the zero function.
3. $f' + f = 0$ gives $f(x) = Ce^{-x}$, therefore $f'(x) = -Ce^{-x}$. The condition relates to the derivative: $f'(1) = 2$ is equivalent to $-Ce^{-1} = 2$, i.e. $C = -2e$. Thus $f(x) = -2e^{1-x}$.